



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE (Main)

FULL TEST

10-01-2023

JEE(Main+Advanced) : ENTHUSIAST COURSE [PHASE : TOAS, TNAS, TAAS, TEAS, TIAS, TMAS, I(A), LIVE-I(A), TRAS, I, LIVE-I, III, IV & V] (SCORE-I)

ANSWER KEY

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	C	A	D	A	B	D	A	C	D
SECTION-II	Q.	11	12	13	14	15	16	17	18	19	20
	A.	B	A	C	C	B	A	B	C	C	A

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	C	A	D	B	A	B	C	C	B
SECTION-II	Q.	11	12	13	14	15	16	17	18	19	20
	A.	B	A	B	B	D	A	D	B	C	A

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	D	C	B	C	C	A	B	B	B
SECTION-II	Q.	11	12	13	14	15	16	17	18	19	20
	A.	C	D	A	C	D	C	A	C	A	C

HINT – SHEET

PART-1 : PHYSICS

SECTION-I

1. Ans (D)

Difference of energy of two consecutive lyman series photon energy will be energy difference of two consecutive energy level. Now only energy difference of 3rd level to 2nd level fall in visible region.

2. Ans (C)

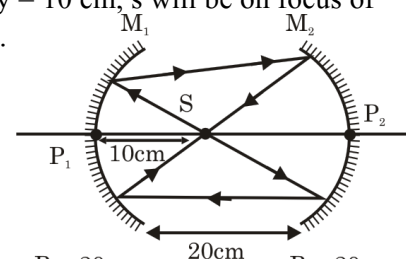
Stopping potential does not depend on intensity of light, material of anode & distance between cathode & anode out of these factors.

3. Ans (A)

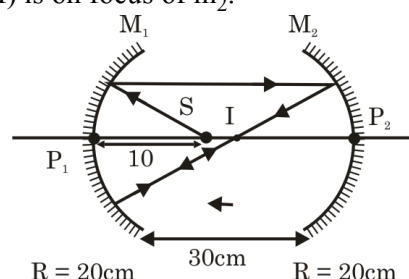
Theoretical, application, uses and properties of Em-waves.

4. Ans (D)

If $x = 20$ cm, $y = 10$ cm, S will be on focus of both m_1 & m_2 .



If $x = 30$ cm, $y = 10$ cm S will be on focus of one mirror and centre of curvature of m_2 . Image of S by m_2 is on S, and image of S by m_1 (I) is on focus of m_2 .



5. **Ans (A)**

with path difference $\frac{\lambda}{2}$, $I_1 = I_0 \cos^2 \left(\frac{\pi}{\lambda} \cdot \frac{\lambda}{2} \right) = 0$

with path diff $\frac{\lambda}{8}$, $I_2 = I_0 \cos^2 \left(\frac{\pi}{\lambda} \cdot \frac{\lambda}{8} \right)$

$$\text{Now } \frac{I_1}{I_2} = 0$$

6. **Ans (B)**

$$\frac{T}{A} = y \cdot \frac{\Delta \ell}{\ell} \Rightarrow f = \frac{1}{2\ell} \sqrt{\frac{T}{\mu}} = \frac{1}{2\ell} \sqrt{\frac{T}{\rho A}}$$

$$\Rightarrow f = \frac{1}{2\ell} \sqrt{\frac{y \cdot \frac{\Delta \ell}{\ell}}{\rho}}$$

$$\Rightarrow f_1 = \frac{1}{2 \times 2} \sqrt{\frac{2.2 \times 10^{11} \times 0.01}{7.7 \times 10^3}}$$

$$f_1 = \frac{1000}{4} \times \sqrt{\frac{2}{7}} = 132.5 \text{ Hz}$$

$$f_2 = 2f_1 = 265 \text{ Hz} \Rightarrow \Delta f = 132.5 \text{ Hz}$$

7. **Ans (D)**

$$F_v = \eta A \cdot \frac{dv}{dr} = \eta \times 2\pi RL \times \frac{\omega R}{d} = \frac{2\pi\eta R^2 \omega L}{d}$$

$$\tau = \frac{2\pi\eta R^2 \omega L}{d} \times R = \frac{2\pi\eta R^3 \omega L}{d}$$

$$= \frac{2 \times 22 \times 0.007 \times 8 \times 10^3 \times 10^{-6} \times 25 \times 40 \times 10^{-2}}{7 \times 5 \times 10^{-4}}$$

$$= 7.04 \text{ N-m}$$

8. **Ans (A)**

$$W_{\text{net}} = -V e \text{ and So } Q_{\text{net}} = -V e$$

$$\Delta U = 0 \text{ for full cycle.}$$

9. **Ans (C)**

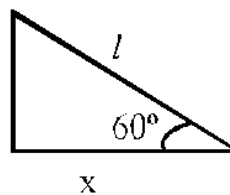
$$\frac{2}{1 - \gamma_{\text{mix}}} - \frac{1}{1 - \frac{5}{3}} + \frac{1}{1 - \frac{7}{5}} = -\frac{3}{2} - \frac{5}{2} = -4$$

$$1 - \gamma_{\text{mix}} = -\frac{1}{2} \Rightarrow \gamma_{\text{mix}} = \frac{3}{2}$$

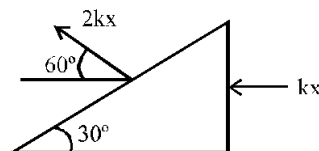
$$T_1 \cdot V_1^{\gamma_{\text{mix}} - 1} = T_2 \cdot V_2^{\gamma_{\text{mix}} - 1}$$

$$\frac{T_2}{T_1} = \left(\frac{V_1}{V_2} \right)^{\gamma_{\text{mix}} - 1} = (5)^{\frac{3}{2} - 1} = (5)^{0.5}$$

10. **Ans (D)**



$$x = \ell \cos 60^\circ \Rightarrow \ell = 2x$$



$$F_{\text{net}} = -(kx + kx) = -2kx$$

$$T = 2\pi \sqrt{\frac{m}{2K}}$$

11. **Ans (B)**

Velocity of two ends of rod Just after collision are

$$V_1 = e_1 V, V_2 = e_2 V \text{ \& } V = \sqrt{2gh}$$

$$\therefore V_{\text{com}} = \left(\frac{V_1 + V_2}{2} \right)^2$$

$$= \frac{(e_1 + e_2)^2}{4} \cdot 2gh = \frac{(e_1 + e_2)^2}{2} gh$$

$$\therefore h_{\text{max}} = \frac{V_{\text{com}}^2}{2g} = \frac{(e_1 + e_2)^2 h}{4}$$

12. **Ans (A)**

Balancing length is measured from P.

$$\text{So } 100 - 49 = 51 \text{ cm}$$

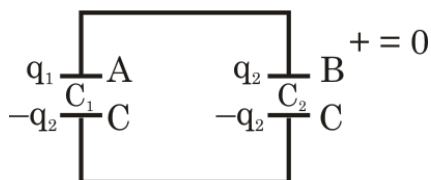
$$E_2 = \phi \times 51$$

Where ϕ = Potential gradient

$$1.02 = \phi \times 51$$

$$\phi = 0.02 \text{ V/cm}$$

13. Ans (C)



Both are 11 to each other with time $C_1 \uparrow$ es &

$C_2 \downarrow$ es

For same potential difference across two capacitor charge on C_1 will $\uparrow$ & that on C_2 will $\downarrow$

$$C_1 = \frac{\epsilon A}{d - ut}, C_2 = \frac{\epsilon_0 A}{d + ut}$$

$$q_{\text{net}} = 2Q$$

$$\therefore \frac{q_1}{q_2} = \frac{C_1}{C_2}$$

$$q_1 = \frac{2QC_1}{C_1 + C_2}$$

$$q_1 = \frac{Q(d + ut)}{d}$$

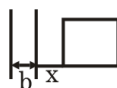
$$\frac{dq_1}{dt} = \frac{Qu}{d}$$

14. Ans (C)

$$\phi = \frac{\mu_0 IL}{2\pi} \ln \left(\frac{a + L}{L} \right) \quad [\text{flux linked with}]$$

loop due to a current carrying wire]

a = distance of nearer edge of wire



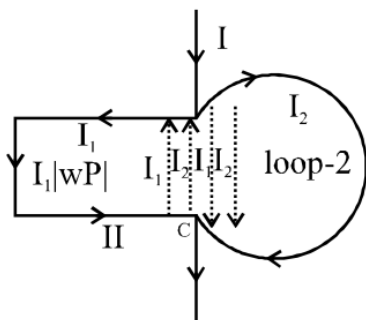
$$\therefore Q_A = Q_B$$

$$\frac{\mu_0 2IL}{2A} \ln 2 = \frac{\mu_0 IL}{2A} \ln \frac{L+x}{x} \Rightarrow x = \frac{L}{3} \therefore b = \frac{2L}{3}$$

15. Ans (B)

$$I = I_1 + I_2$$

$$\vec{F}_{\text{loop 1}} = 0, \vec{F}_{\text{loop 2}} = 0$$



$$\text{For Additional wire } |\vec{F}| = (I_1 + I_2) B \ell$$

$$|\vec{F}| = IB \ell$$

16. Ans (A)

Conceptual

17. Ans (B)

$$V = \sqrt{\frac{Gm}{r}}$$

$$TE = -\frac{GMm}{2r} = -E$$

so the energy given to satellite to escape it from gravity influence of earth is E .

so total energy would become zero.

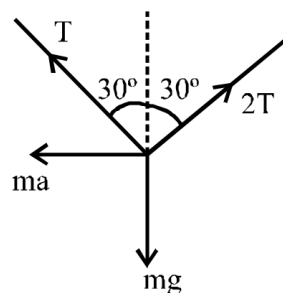
so change in kinetic energy

$$\Delta E = E \text{ \& } 2E = \frac{1}{2}mv_f^2 \Rightarrow v_f = \sqrt{\frac{4E}{m}}$$

$$\text{now } \Delta V = (\sqrt{2} - 1) 2$$

18. Ans (C)

inframed cart



$$\therefore (2T + T) \cos 30 = mg$$

$$T = \frac{2mg}{3\sqrt{3}}$$

$$ma + T \sin 30 = 2T \sin 30$$

$$ma + T \sin 30$$

$$ma = \frac{mq}{3\sqrt{3}}$$

$$a = \frac{g}{3\sqrt{3}}$$

19. Ans (C)

$$\frac{B}{E} = \frac{1}{c} \text{ \& } \mu_0 \epsilon_0 = \frac{1}{c^2}$$

$$\therefore \frac{B}{E} \cdot \mu_0 \epsilon_0 = \frac{1}{c^3} = M^0 L^{-3} T^3$$

20. Ans (A)

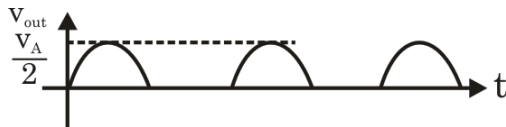
Percentage error is least in Case-1

PART-1 : PHYSICS

SECTION-II

1. **Ans (5)**

Output will be half wave rectified and of half the amplitude.



$$V_{avg} = \frac{V_{peak}}{\pi} = \frac{V_0}{2\pi} = \frac{10\pi}{2\pi} = 5$$

$$V_{avg} = \frac{1}{T} \int_0^T V_{out} dt$$

$$= \frac{1}{T} \left[\int_0^{T/2} V_0 \sin \omega t dt + \int_{T/2}^T 0 dt \right] = \frac{V_0}{\pi}$$

2. **Ans (2)**

$$I_{out} = \left(\frac{I_0}{2} \right) \cos^2(45) \times \cos^2(45) \cos^2(45)$$

$$I_{out} = \frac{I_0}{2^4} = \frac{I_0}{16} = 2 \text{ W/m}^2$$

3. **Ans (5)**

$$\omega = \sqrt{\omega_0^2 - \gamma^2}$$

$$= \omega_0 \left(1 - \frac{\gamma^2}{2\omega_0^2} \right) = \omega_0 - \frac{\gamma^2}{2\omega_0}$$

$$\omega_0 - \omega = \frac{\gamma^2}{2\omega_0} = \frac{\omega_0 - \omega}{\omega_0}$$

$$= \frac{\gamma^2}{2\omega_0^2} = \frac{1}{2} \times 10^{-4} = 5 \times 10^{-3} \%$$

4. **Ans (2)**

$$f_0 = \frac{R_0}{2}, f = \frac{R}{2} = \frac{R_0(1 + \alpha \Delta T)}{2}$$

$$f = f_0(1 + \alpha \Delta T)$$

$$\Rightarrow \frac{f - f_0}{f_0} = \alpha \Delta T = 10^{-3} \times 20 = 2\%$$

5. **Ans (20)**

Process 1 → 2 and 3 → 4 are isochoric process. So work done in these two process would be zero.

$$W_{23} = P \Delta V = nR \Delta T = 4800 \text{ R}$$

$$W_{41} = P \Delta V = nR \Delta T = -2400 \text{ R}$$

$$W_{net} = 2400 \times \frac{25}{3} = 20 \text{ kJ}$$

6. **Ans (5)**

Let impedance of circuit = Z

impedance of box = x + yi (i = √-1)

$$\therefore Z = (x + yi) + \omega Li$$

$$Z = x + (y + \omega L) i$$

If power factor of circuit = 1

$$\Rightarrow y + \omega L = 0, y = -\omega L = -100 \times .01 = -10 \Omega$$

$$I = \text{current in circuit} \frac{V_{rms}}{x} \Rightarrow x = \frac{25}{5} = 5 \Omega$$

∴ Cos ϕ for box

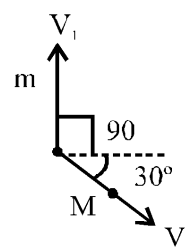
$$= \frac{x}{\sqrt{x^2 + y^2}} = \frac{5}{\sqrt{5^2 + 10^2}} = \frac{1}{\sqrt{5}} \Rightarrow n = 5$$

7. **Ans (9)**

$$E_{eq} = \frac{\frac{4}{6} + \frac{12}{3} + \frac{8}{6}}{\frac{1}{6} + \frac{1}{3} + \frac{1}{6}}$$

$$= \frac{4 + 24 + 8}{4} = 1 + 6 + 2 = 9 \text{ volt}$$

8. **Ans (5)**



before collision

By COLM along line of motion of m

$$mu = mv_2 \cos 30$$

$$mu = mv_2 \frac{\sqrt{3}}{2} \dots (i)$$

$$v_2 = \frac{m}{M} \frac{2u}{\sqrt{3}}$$

By COLM along + ar to line of motion of m

$$0 = mv_1 - Mv_2 \sin 30 \dots (ii)$$

$$v_1 = \frac{v_2}{2} \frac{M}{m}$$

$$K_1 = \frac{1}{2} mu^2$$

$$K_f = 60\% \text{ of } K_i$$

$$\frac{1}{2} MV_2^2 + \frac{1}{2} MV_1^2 = \frac{3}{5} \frac{1}{2} mu^2$$

$$\frac{1}{2} mV_2^2 = \left(\frac{3}{5} - \frac{1}{3} \right) \frac{1}{2} mu^2 = \frac{4}{3 \times 5} - \left(\frac{1}{2} mu^2 \right)$$

$$\frac{1}{2} M \times \left(\frac{M}{m} \right)^2 \times \frac{4u^2}{3} = \frac{4 \times 1 mu^2}{5 \times 2 \times 3}$$

9. **Ans (5)**

By COLM

$$P_{\text{sys}} = (75 + 25) 4 = 75 (V_p + 2) + 25V_p$$

$$400 = 100V_p + 150$$

$$V_p = 2.5 \text{ m/s}$$

$$d = 2.5 \times \frac{4}{2} = 5 \text{ m}$$

10. **Ans (9)**

By COML

$$\therefore mgh = \frac{1}{2}mv^2$$

$$h = \frac{v^2}{2g}$$

$$\frac{x}{5} = \frac{36}{2 \times 10} = 1.8$$

(Rotational K.E. remain same)

PART-2 : CHEMISTRY

SECTION-I

1. **Ans (B)**

$$l = \frac{h}{mv} \text{ Now } v \propto \left(\frac{Z}{n}\right) \text{ so } l \propto \left(\frac{n}{Z}\right)$$

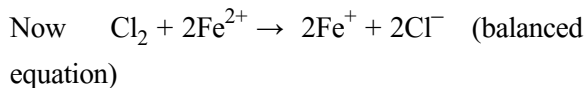
$$\text{for second orbit of } \text{Li}^{2+} l \propto \left(\frac{2}{3}\right)$$

$$\text{for } n = 4 \text{ of } \text{C}^{5+} \text{ ion, } l \propto \left(\frac{4}{6}\right) = 2/3 \text{ Hence the result}$$

2. **Ans (C)**

$$\begin{aligned} \text{Excess of Mohr's salt} &= 30 \times 0.1 \times 5 \text{ meq.} \\ &= 15 \text{ meq} \end{aligned}$$

$$\begin{aligned} \text{Mohr's salt consumed} &= [(35 \times 1.0 \times 1) - 15] \text{ meq} \\ &= 20 \text{ meq.} \end{aligned}$$



$$\begin{aligned} \text{So moles of } \text{Cl}_2 &= 1/2 \times \text{moles of } \text{Fe}^{2+} \\ &= 1/2 \text{ eq. of } \text{Fe}^{2+} = 1/2 \times 20 \text{ m moles} = 10 \text{ moles} \\ &= 10 \times 10^{-3} \times 71 \text{ gm} = 0.71 \text{ gm} \end{aligned}$$

$$\text{So, required \% } \frac{0.71}{10} \times 100 = 7.1\%$$

3. **Ans (A)**

Shown arrangement is hexagonally closed pack plane & these type of planes are arranged perpendicular to body diagonal of fcc unit cell as shown.

4. **Ans (D)**

Electrophoresis means movement of colloidal particles under the influence of electric field.

5. **Ans (B)**

$$\Delta S = nC_V \ln \left(\frac{T_f}{T_i}\right) + nR \ln \left(\frac{V_f}{V_i}\right)$$

6. **Ans (A)**

Slope of adiabatic curve is greater than isothermal curve

8. **Ans (C)**

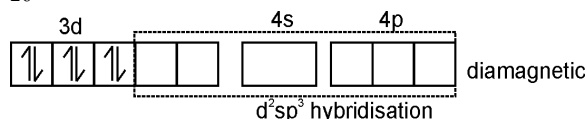
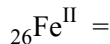
(A) $[\text{Pt}(\text{en})(\text{SCN})_2(\text{NH}_3)_2]^0$; it is not ion

(B) $[\text{Pt}(\text{en})(\text{SCN})_2(\text{NH}_3)_2]^{2+}$; it should not be platinate

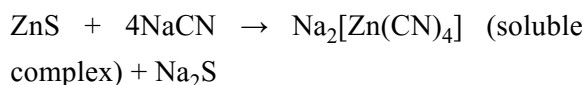
(C) correct name

(D) $[\text{Pt}(\text{en})_2(\text{SCN})_2(\text{NH}_3)_2]^{2+}$; its co-ordination number = 8

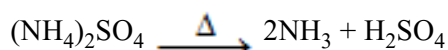
9. **Ans (C)**



10. **Ans (B)**



11. **Ans (B)**



12. **Ans (A)**

(A) $\text{SO}_2 + 2\text{H}_2\text{O} \rightarrow \text{H}_2\text{SO}_4 + 2\text{H}$. Hence bleaching is through reduction (temporary) and $\text{Cl}_2 + \text{H}_2\text{O} \rightarrow 2\text{HCl} + [\text{O}]$ bleaching is through oxidation (permanent).

So (A) is correct and (B), (C) and (D) are incorrect.

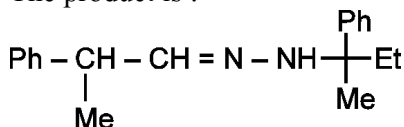
13. **Ans (B)**

(A), (C) and (D) are correct statements

(B) First decreases from B to Al and then increases marginally owing to discrepancies in atomic size of the element.

14. **Ans (B)**

The product is :



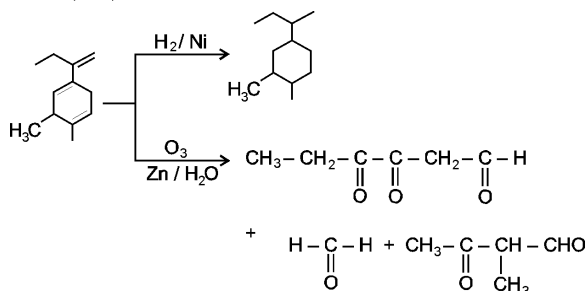
I = (+, +) (E)

II = (+, +) (Z)

III = (-, +) (E)

IV = (-, +) (Z)

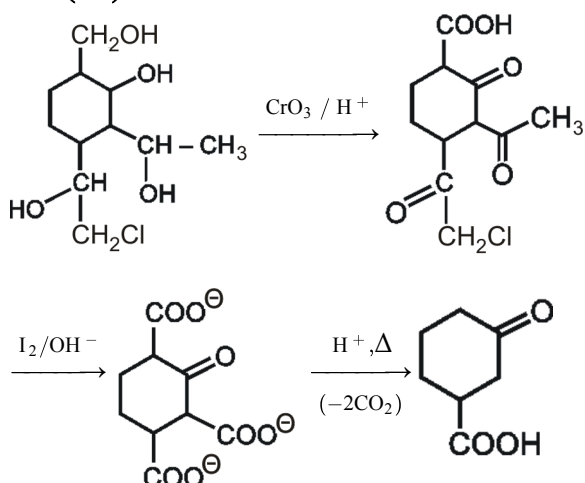
15. **Ans (D)**



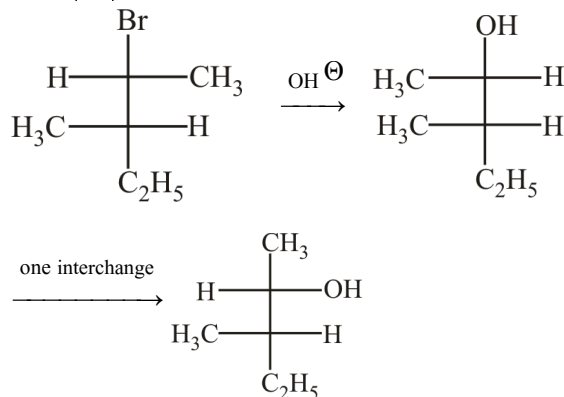
16. **Ans (A)**

Given compound is 'S' in configuration due to Walden inversion on attack by IQ we get 'R' configuration.

17. **Ans (D)**



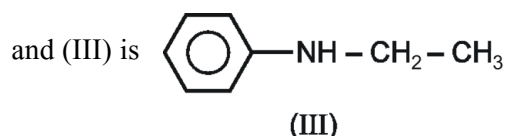
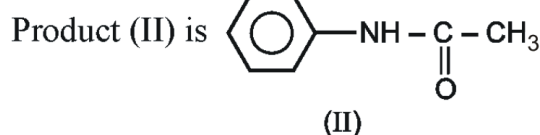
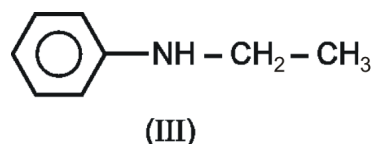
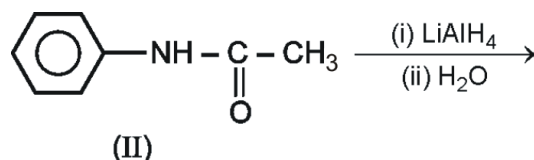
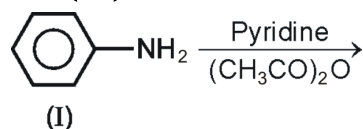
18. **Ans (B)**



19. **Ans (C)**

Intramolecular aldol condensation take place.

20. **Ans (A)**



$\therefore \text{III} > \text{I} > \text{II}$ as + I effect of increases the basic strength and $-\text{COCH}_3$ by $-\text{R}$, $-\text{I}$ reduces the basic strength.

PART-2 : CHEMISTRY

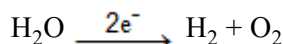
SECTION-II

1. **Ans (2)**

$t_{1/2} \propto [\text{Here } a \propto P]$

$$\frac{880}{410} = \left[\frac{364}{170} \right]^{n-1} \therefore n = 2$$

2. **Ans (1930)**



$$2 \times 96500 \text{ C} \rightarrow 22.4 \times 1000 \text{ cm}^3 \text{ H}_2$$

$$\frac{2 \times 96500 \times 224}{22.4 \times 1000} \text{ C} \leftarrow \therefore 224 \text{ cm}^3 \text{ H}_2$$

$$= 1930 \text{ C}$$

3. **Ans (67)**

$$V_A : V_B : V_C = \frac{1}{50} : \frac{1}{100} : \frac{1}{150} = 6 : 3 : 2$$

$$\frac{1}{50} = \Lambda_m C_A \left(\frac{A}{\ell} \right) \dots\dots\dots (1)$$

$$\frac{1}{100} = \Lambda_m C_B \left(\frac{A}{\ell} \right)$$

$$\frac{1}{150} = \Lambda_m C_C \left(\frac{A}{\ell} \right)$$

$$C_A : C_B : C_C = 6 : 3 : 2$$

$$\frac{1}{R} = \Lambda_m \left(\frac{C_A V_A + C_B V_B + C_C V_C}{V_A + V_B + V_C} \right) \left(\frac{A}{e} \right)$$

$$\dots\dots\dots (2)$$

Divide (1) / (2) :

$$\frac{R}{50} = \frac{C_A [V_A + V_B + V_C]}{C_A V_B + C_B V_B + C_C V_C} = \frac{1 + \frac{V_B}{V_A} + \frac{V_C}{V_A}}{1 + \frac{C_B}{C_A} \cdot \frac{V_B}{V_A} + \frac{C_C}{C_A} \cdot \frac{V_C}{V_A}}$$

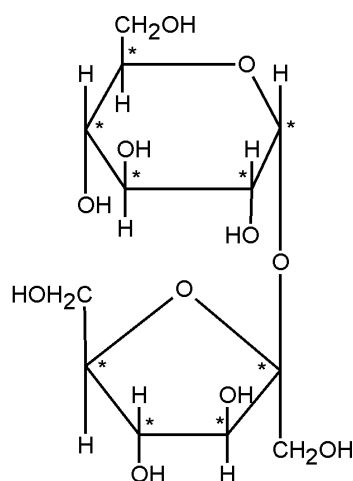
$$= \frac{18 + 9 + 6}{18 + 4.5 + 2} = \frac{33}{24.5}$$

$$\Rightarrow R = 67.34$$

5. **Ans (3)**

$m = 1.73$ i.e. one unpaired electron, Ti (E.C.) = $[\text{Ar}]^{18} 3d^2, 4s^2$, so it will lose three electrons to have one unpaired electron and thus the oxidation state of titanium is +3

8. **Ans (9)**



The star shown indicates chiral centre.

PART-3 : MATHEMATICS

SECTION-I

1. **Ans (D)**

$$(A) P \wedge (P \vee Q) \equiv P$$

$$(B) P \wedge (P \wedge Q) \equiv P \wedge Q$$

$$(C) Q \rightarrow (P \wedge (P \rightarrow Q))$$

$$\equiv Q \rightarrow (P \wedge (\sim P \vee Q)) \equiv Q \rightarrow (P \wedge Q)$$

$$\equiv (\sim Q) \vee (P \wedge Q) \equiv (P \vee (\sim Q))$$

(D) Make truth table

2. **Ans (D)**

Mean

$$= \frac{6 + 10 + 7 + 13 + a + 12 + b + 12}{8} = 9$$

$$60 + a + b = 72$$

$$a + b = 12 \dots\dots(1)$$

$$\text{variance} = \frac{\sum x_i^2}{n} - \left(\frac{\sum x_i}{n} \right)^2 = \frac{37}{4}$$

$$\sum x_i^2 = 6^2 + 10^2 + 7^2 + 13^2 + a^2 + b^2 + 12^2 + 12^2$$

$$= a^2 + b^2 + 642$$

$$\frac{a^2 + b^2 + 642}{8} - (9)^2 = \frac{37}{4}$$

$$\frac{a^2 + b^2}{8} + \frac{321}{4} - 81 = \frac{37}{4}$$

$$\frac{a^2 + b^2}{8} = 81 + \frac{37}{4} - \frac{321}{4}$$

$$\frac{a^2 + b^2}{8} = 81 - 71$$

$$\therefore a^2 + b^2 = 80 \dots\dots(2)$$

$$\text{From (1) } a^2 + b^2 + 2ab = 144$$

$$80 + 2ab = 144 \therefore 2ab = 64$$

3. **Ans (C)**

$$2PA = 3PB$$

$$\begin{aligned} &\Rightarrow 2\sqrt{(x-9\sqrt{2})^2 + (y-9\sqrt{2})^2} \\ &= 3\sqrt{(x-4\sqrt{2})^2 + (y+4\sqrt{2})^2} \\ &\Rightarrow 4[x^2 + y^2 - 18\sqrt{2}x - 18\sqrt{2}y + 324] \\ &= 9[x^2 + y^2 - 8\sqrt{2}x - 8\sqrt{2}y + 64] \\ &\Rightarrow 5(x^2 + y^2) = 720 \\ &\Rightarrow x^2 + y^2 = 144 \end{aligned}$$

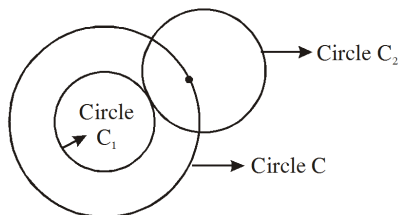
Let this circle be C_1 , then circle C_2 of radius 3 touches C_1 externally.

$\therefore$ locus of centre of circle C_2 will also be a circle 'C' concentric

with circle C_1 and of radius = radius of circle $C_1 + 3 = 12 + 3$

$\therefore$ locus of centre of circle C_2

i.e. equation of circle 'C' is $x^2 + y^2 = 15^2$

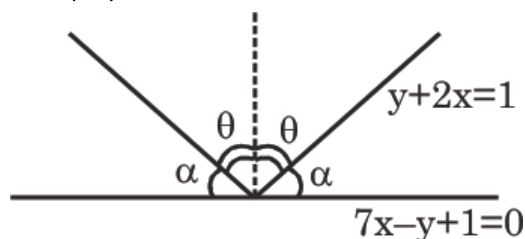


4. **Ans (B)**

As the line joining the mid point of the chord joining (1, 2), (3, 4) and the point of intersection of tangents at these points is parallel to axis of parabola

$$\therefore \text{Slope of axis} = \frac{8-3}{-6-2} = -\frac{5}{8}$$

5. **Ans (C)**



$\Rightarrow$ Equation of incident ray is

$$41x - 38y + 38 = 0$$

6. **Ans (C)**

$$\alpha_1 \cdot \beta_1 = \sqrt{17}, \alpha_2 \cdot \beta_2 = \sqrt{17}$$

$$\text{Let } P \left(\alpha_1, \frac{\sqrt{17}}{\alpha_1} \right) \text{ and } Q \left(\alpha_2, \frac{\sqrt{17}}{\alpha_2} \right)$$

P and Q lie on director circle of ellipse

$$x^2 + y^2 = 10 \Rightarrow \alpha_1^2 + \frac{17}{\alpha_1^2} = 10$$

$$\Rightarrow \alpha_1^4 - 10\alpha_1^2 + 17 = 0$$

$$\Rightarrow \alpha_1^2 = \frac{10 \pm \sqrt{100 - 68}}{2} = 5 \pm 2\sqrt{2}$$

$$\alpha_1 = \pm \sqrt{5 \pm 2\sqrt{2}}$$

As P and Q are in I quad.

$$\alpha_1 = \sqrt{5 + 2\sqrt{2}}, \alpha_2 = \sqrt{5 - 2\sqrt{2}}$$

$$\Rightarrow \alpha_1^2 - \alpha_2^2 = 5 + 2\sqrt{2} - 5 + \sqrt{2}$$

$$= 4\sqrt{2} = \sqrt{32} \quad [\alpha_1^2 - \alpha_2^2] = 5$$

7. **Ans (A)**

$$\frac{dy}{dx} = 2x + 1 \Rightarrow \int dy = \int (2x + 1) dx + c$$

$$\Rightarrow y = x^2 + x + c$$

Curve passes through

$$(1, 2) = 2 = 1 + 1 + c \Rightarrow c = 0$$

$$y = x^2 + x$$

Required area =

$$\int_0^1 (x^2 + x) dx = \left(\frac{x^3}{3} + \frac{x^2}{2} \right)_0^1 = \frac{1}{3} + \frac{1}{2} = \frac{5}{6}$$

8. **Ans (B)**

$$5^{\frac{1}{2} \log_7 (49+20\sqrt{6})} \cdot \frac{1}{2} \log_5 7$$

$$= 5^{\frac{1}{4} \frac{\log_7 (49+20\sqrt{6})}{\log_7 5}}$$

$$= 5^{\log_5 (49+20\sqrt{6})^{\frac{1}{4}}}$$

$$= (49 + 20\sqrt{6})^{\frac{1}{4}}$$

$$= (5 + 2\sqrt{6})^{\frac{1}{2}} = (5 + 2\sqrt{6})^{\frac{1}{2}}$$

$$= \sqrt{3} + \sqrt{2}$$

9. **Ans (B)**

Equation of tangent

$$y = mx \pm \sqrt{m^2 - 1}$$

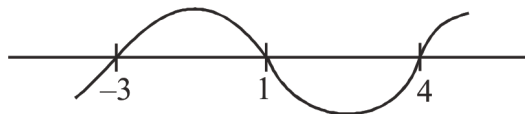
$$m = 2 \Rightarrow y = 2x \pm \sqrt{4 - 1}$$

$$= c = \pm \sqrt{3}$$

10. **Ans (B)**

$$f(x) = (x-1)(x+3)(x+4)$$

$$f(x) = 0, \Rightarrow x = 1, -3, 4$$



$x = 1 \Rightarrow$ point of local maxima

$x = -3, x = 4$ is point of local minima

$$\alpha_1 = 4, \alpha_2 = 1, \alpha_3 = 3$$

(Q) & (R) statement are true.

11. **Ans (C)**

$$\left| \frac{z-i}{z+2i} \right| = 1$$

$$\Rightarrow |z-i| = |z+2i|$$

$\Rightarrow z$ lies on perpendicular bisector of $(0, 1)$ and $(0, -2)$.

$$\Rightarrow \text{Im}z = -\frac{1}{2}$$

$$\text{Let } z = x - \frac{i}{2}$$

$$\because |z| = \frac{5}{2} \Rightarrow x^2 = 6$$

$$\begin{aligned} \therefore |z+3i| &= \left| x + \frac{5i}{2} \right| = \sqrt{x^2 + \frac{25}{4}} \\ &= \sqrt{6 + \frac{25}{4}} = \frac{7}{2} \end{aligned}$$

12. **Ans (D)**

$$\begin{aligned} S &= 3 \left(\frac{1}{2} + \frac{1}{6} + \frac{1}{12} + \frac{1}{20} + \dots \right) \\ &= 3 \left(\left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \left(\frac{1}{3} - \frac{1}{4}\right) + \left(\frac{1}{4} - \frac{1}{5}\right) + \dots \right) \\ S &= 3 \end{aligned}$$

13. **Ans (A)**

$$P_n - 7P_{n-1} - 3P_{n-2} = 0$$

14. **Ans (C)**

$$xyz = 2^4 \times 3^1$$

$$\text{Let } x = 2^{\alpha_1} \times 3^{\beta_1}$$

$$y = 2^{\alpha_2} \times 3^{\beta_2}$$

$$z = 2^{\alpha_3} \times 3^{\beta_3}$$

$$\text{Now } \alpha_1 + \alpha_2 + \alpha_3 = 4$$

$$\text{No. of non-negative integral sol} = {}^6C_2 = 15$$

$$\& \beta_1 + \beta_2 + \beta_3 = 1$$

$$\text{No. of non-negative integral sol}^n = {}^3C_2 = 3$$

$$\text{Total ways} = 15 \times 3 = 45.$$

15. **Ans (D)**

All throws are independent

$$\therefore P = \frac{1}{6}$$

16. **Ans (C)**

Given that

$$f(x) = |1 - 2x|,$$

$$\Rightarrow f\left(\left|\frac{x}{2}\right|\right) = \begin{cases} x-1, & x > 1 \\ 1-x, & 0 < x \leq 1 \\ 1+x, & -1 \leq x \leq 0 \\ -x-1, & x < -1 \end{cases}$$

Clearly the domain of $\sin^{-1}\left(f\left|\frac{x}{2}\right|\right)$ is $[-2, 2]$.

$\Rightarrow$ It is non differentiable at the points $\{-1, 0, 1\}$.

17. **Ans (A)**

$$\int e^{\sin x} \left(\frac{5 + \cos^2 x}{(2 - \sin x) \sqrt{3 + \cos^2 x}} \right) \cos x \, dx$$

put $\sin x = t$

$$\int e^t \left(\frac{6 - t^2}{(2 - t) \sqrt{4 - t^2}} \right) dt$$

$$\int e^t \left(\frac{4 - t^2}{(2 - t) \sqrt{4 - t^2}} + \frac{2}{(2 - t) \sqrt{4 - t^2}} \right) dt$$

$$\int e^t \left(\sqrt{\frac{2+t}{2-t}} + \frac{2}{(2-t)^{3/2}(2+t)^{1/2}} \right) dt$$

$$\text{If } g(t) = \sqrt{\frac{2+t}{2-t}},$$

$$g'(t) = \frac{2}{(2-t)^{3/2}(2+t)^{1/2}}$$

$$e^t \sqrt{\frac{2+t}{2-t}} + c$$

$$f(x) = e^{\sin x} \sqrt{\frac{2 + \sin x}{2 - \sin x}} + c$$

$$f(0) = 1 \Rightarrow c = 0$$

$$f(x) = e^{\sin x} \sqrt{\frac{2 + \sin x}{2 - \sin x}}$$

$$f\left(\frac{\pi}{2}\right) = \sqrt{3}e$$

18. **Ans (C)**

$$\vec{c} \cdot (\vec{a} \times \vec{b}) = 0$$

$$\Rightarrow \begin{vmatrix} 1 & -2 & 3 \\ 3 & 3 & -1 \\ d & 1 & 2d-1 \end{vmatrix} = 0 \Rightarrow 11d = -1$$

19. **Ans (A)**

Required plane is

$$[r - (2\hat{i} + \hat{j} - 3\hat{k})] \cdot (2\hat{i} + \hat{j} - 3\hat{k}) = 0$$

20. **Ans (C)**

$$2A + B = B^T - A^T \quad \dots(i)$$

take transpose both side

$$2A^T + B^T = B - A \quad \dots(ii)$$

equation (i) - equation (ii)

$$\Rightarrow 3A = -3A^T \Rightarrow A^T = -A$$

$$\Rightarrow 2A + B = B^T + A$$

$$\Rightarrow A + B = B^T$$

$$\Rightarrow |A + B| = |B^T| = |B| = 1$$

PART-3 : MATHEMATICS

SECTION-II

1. **Ans (1)**

Put $z = x + iy$

$$(x + iy) + i(|z|) = i(x - iy) + 1$$

$$\Rightarrow x = y + 1 \text{ \& } y + |z| = x$$

$$\Rightarrow x - y = 1 \text{ \& } x - y = |z|$$

$$\Rightarrow |z| = 1$$

2. **Ans (3)**

$$\lim_{x \rightarrow \infty} \frac{e^2 \left[e^{x \ln \left(\frac{x+1}{x-1} \right) - 2} - 1 \right] \left(x \ln \left(\frac{x+1}{x-1} \right) - 2 \right)}{\left(\frac{1}{x^2} \right) \left(x \ln \left(\frac{x+1}{x-1} \right) - 2 \right)}$$

$$L = \frac{2}{3}e^2$$

3. **Ans (3)**

$$g(x) = f^{-1}(x) \Rightarrow g(f(x)) = x$$

$$g'(f(x)) \cdot f'(x) = 1 \Rightarrow g'(f(x)) = \frac{1}{f'(x)}$$

$$f\left(\frac{5}{2}\right) = 3 \Rightarrow g'(f(5/2)) = \frac{1}{f'(5/2)} \Rightarrow g'(3) = \frac{1}{f'(5/2)}$$

$$\Rightarrow f'(x) = \frac{4}{3}(2x-5)^{1/3} \times 2 + \frac{4}{3}(2x^2-3x-5)^{1/3}(4x-3) + \frac{4}{2\sqrt{4x-1}}$$

$$f'(5/2) = 0 + 0 + \frac{2}{3}$$

$$\therefore g'(3) = \frac{3}{2} \Rightarrow 2g'(3) = 3$$

4. **Ans (9)**

$$\sin(\cos x) - \frac{1}{2} \geq 0$$

$$\sin(\cos x) \geq \frac{1}{2}$$

$$\frac{\pi}{6} \leq \cos x \leq 1$$

$$\alpha = \frac{\pi}{6}, 3\cot^2 \frac{\pi}{6} = 9$$

5. **Ans (2)**

Eliminating t will give

$$y^2(x-1) = 1$$

equation of tangent at $(2, 1)$ is $x + 2y = 4$

solving with curve,

$$\text{we get } x = 5 \text{ and } y = -\frac{1}{2}$$

$$Q = \left(5, -\frac{1}{2}\right) \text{ or } |PQ| = \frac{3\sqrt{5}}{2}$$

6. **Ans (27)**

Put $y = 9$ in given pair of straight lines

$$18x^2 - 81x + 81 = 0$$

$$\Rightarrow x = 3, \frac{3}{2}$$

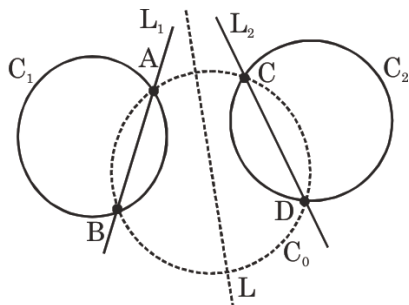
So, we have three vertices $(0, 0)$, $(3, 9)$ and

$$\left(\frac{3}{2}, 9\right)$$

$$\text{Hence } \Delta = \frac{27}{4}$$

7. **Ans (4)**

We represent the given situation through the following diagram :



- C_0 is the circle passing through A, B, C and D.
- L is the radical axis of C_1 and C_2 given by.

$$S_1 - S_2 = 0$$

$$\Rightarrow 2x - 2y - 6 = 0$$

We use the fact that the radical axes of three circles taken two at a time are concurrent. Thus,

L, L_1 and L_2 are concurrent :

$$\begin{vmatrix} 2 & -2 & -6 \\ 1 & 2 & 3 \\ 2 & 3 & \lambda \end{vmatrix} = 0 \Rightarrow \lambda = 4$$

8. **Ans (1)**

$$\lim_{x \rightarrow 0} \left((a-n)n - \frac{\tan x}{x} \right) \frac{\sin nx}{x} = 0$$

$$\Rightarrow ((a-n)n - 1)n = 0$$

$$\Rightarrow a = n + \frac{1}{n}$$

9. **Ans (0)**

Do yourself.

10. **Ans (3)**

$$y = \frac{x^4 - x^2 - 1 - 2x}{x^2 - x - 1}$$

$$y = \frac{(x^2 + x + 1)(x^2 - x + 1)}{(x^2 - x - 1)}$$

$$\Rightarrow y = x^2 + x + 1$$